

Elizaveta Rebrova

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Research interests

Study of the structure and properties of large random matrices (non-asymptotic random matrix theory)
Development and analysis of the algorithms for large and high dimensional data sets
Mathematical data science and high dimensional data analysis
High-dimensional probability, geometric analysis, and computational mathematics as tools for the listed above

Education

2013-2018 **PhD in Mathematics**
University of Michigan, Ann Arbor
Advisor *Roman Vershynin*
Thesis *Spectral properties of heavy-tailed random matrices*
2007-2012 **Specialist degree in Mathematics** (B.S.+M.S. equivalent)
Lomonosov Moscow State University
Diploma with honors.
Master thesis *Sobolev classes on infinite dimensional spaces*.
Advisor *Vladimir I. Bogachev*
2011-2012 **M.S. equivalent in Computer Science**
Yandex School of Data Analysis, Moscow
12 Masters-level courses, including Algorithms and Data Structures with C++, Complexity Theory, Machine Learning, Computational Linguistics, Game Theory, Applied Statistics

Employment

2018-now **Assistant Adjunct Professor**, Applied Mathematics
University of California, Los Angeles
Department of Mathematics
Summer 2018 **Engineer I**
Lawrence Berkeley National Lab, Berkeley, CA
Computational Research Division

Internship experience

Summer 2017 **Lawrence Berkeley National Lab, Berkeley, CA**
Computational Research Division, Scalable Solvers Group;
Internship mentor: Xiaoye Sherry Li
Investigated applications of STRUMPACK linear solver to kernel matrices

Teaching and service

2019-now	Co-mentoring PhD student Will Swartworth on the project "Median method for the corruptions detection" (with D.Needell and J.Haddock)
2018-now	Co-organizing of the Random matrices seminar at UCLA (with P.Salanevich)
2018-now	Professor , University of California, Los Angeles Probability I (Fall 2018, Spring 2020); Statistics (Winter, Spring 2019, Winter 2020); Stochastic Processes (Spring 2020)
2014-2017	Graduate Student Instructor , University of Michigan, Ann Arbor Calculus I (Fall 2014); Differential Equations (Winter 2015, Winter 2016, Fall 2016); Computational Finance (Fall 2017)
2013-2014	Instructor in Math Lab, University of Michigan, Ann Arbor
2012-2013	Algebra instructor , Kolmogorov math and physics high school, Moscow, Russia
2009-2012	Seminar work with math high school students, Moscow State School N 57, Moscow, Russia.
2007-2012	Member of Organizing Committee , Moscow Mathematical and Linguistics Olympiads, Moscow, Russia.

Fellowships, grants, awards

2018	Capital Fund Management sponsored postdoc grant
2018	Allen Shields Memorial Fellowship, University of Michigan
2017	NSF scholarship for summer research internship at LBNL
2017	Rackham One-Term Dissertation Fellowship, University of Michigan
2014-2016	University of Michigan Math Department Graduate Summer Fellowship
2012	RFFI 10-01-00518, research grant, Russia
2010	Excellent progress mayor's scholarship, Moscow, Russia
2007	The 5th International Linguistics Olympiad (for high school students), 1st place team award and individual Bronze award

Technical skills

Math Software	Matlab, Mathematica, Excel/Sheets, LATEX
Languages	Python, C/C++
ML related	jupyter, pandas, numpy, scikit-learn

Publications and preprints

Preprints

- (2) M. Iwen, D. Needell, E. Rebrova, A. Zare, Lower Memory Oblivious (Tensor) Subspace Embeddings with Fewer Random Bits: Modewise Methods for Least Squares, arXiv:1912.08294
- (1) D. Needell, E. Rebrova, On block Gaussian sketching for iterative projections, arXiv:1905.08894

Journal articles

- (5) E. Rebrova, Constructive local regularization of the operator norm of random matrices, *J. Theor. Probab.*, to appear

- (4) E. Rebrova, R. Vershynin, Norms of random matrices: local and global problems, *Advances in Mathematics*, Vol. 324, pp 40–83 (2018)
- (3) E. Rebrova, K. Tikhomirov, Coverings of random ellipsoids, and invertibility of matrices with i.i.d. heavy-tailed entries, *Israel J. Math.*, Vol. 227(2), pp 507-544 (2018)
- (2) V.I. Bogachev, A.Y. Pilipenko, E. Rebrova, Classes of functions of bounded variation on infinite-dimensional domains, *Doklady Mathematics* (Doklady Akademii Nauk), 88(1), pp 391-395 (2013)
- (1) V.I. Bogachev, E. Rebrova, Functions of bounded variation on infinite-dimensional spaces with measures, *Doklady Mathematics* (Doklady Akademii Nauk), 87(2), pp 144-147 (2013)

Conference publications

- (3) E. Rebrova, D. Needell, Sketching for Motzkin’s iterative method for linear systems, Proc. 50th Asilomar Conf. on Signals, Systems and Computers, Pacific Grove, CA (2019)
- (2) E. Rebrova, D. Needell, New bounds for the block Gaussian sketch and project method, Proc. Information Theory and Applications, La Jolla CA (2019)
- (1) E. Rebrova, G. Chavez, Y. Liu, P. Ghysels and X.S.Li, A study of clustering techniques and hierarchical matrix formats for kernel ridge regression, 2018 IEEE IPDPS ParLearning workshop (2018)

Invited talks

- November 2019 *Modewise Johnson-Lindenstrauss embeddings for tensors*, Combinatorics and Probability seminar, University of California, Irvine, CA
- November 2019 *Sketching for Motzkin’s iterative method*, Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA
- October 2019 *Sketching for iterative linear solvers: Strengths, weaknesses, and tools from random matrix theory*, Probability seminar, Stanford, CA
- October 2019 *Sketching for iterative linear solvers: Strengths, weaknesses, and tools from random matrix theory*, Probability seminar, University of Southern California, CA
- March 2019 *Iterative linear solvers and random matrices: new bounds for the block Gaussian sketch and project method*, High-dimensional seminar, GeorgiaTech, Atlanta, GA
- March 2019 *Constructive regularization of random matrix norm*, Probability seminar, University of Alberta, Edmonton, Canada
- February 2019 *Iterative linear solvers and random matrices: new bounds for the block Gaussian sketch and project method*, Southern California Applied Mathematics Symposium, Caltech, Pasadena, CA
- September 2018 *Constructive regularization of random matrix norm*, Structural Inference in High-Dimensional Models workshop, HSE, Moscow, Russia
- August 2018 *Constructive regularization of random matrix norm*, SUMIFRAS workshop, TAMU University, College Station, TX
- June, 2017 *Non-asymptotic spectral properties of heavy-tailed random matrices*, two talks, Université-Paris-Est Marne-la-Vallée and Université Paris Diderot, Paris, France

May, 2017	<i>Norms of heavy-tailed random matrices: local and global problems</i> , University of Alberta, Edmonton, Canada
March, 2017	<i>Local and global obstructions for the random matrix norm regularization</i> , MIC seminar at Center for Data Science, NYU, New York
June, 2016	<i>Regularization of the norm of matrices with i.i.d. heavy-tailed entries</i> , CMS Summer Sectional Meeting University of Alberta, Edmonton, Canada
March, 2016	<i>Coverings of random ellipsoids, and invertibility of matrices with i.i.d. heavy-tailed entries</i> , AMS Sectional Meeting, University of Georgia, Athens, GA

Other talks

October, 2019	<i>Eigenvectors of random matrices: a survey by Sean O'Rourke, Van Vu, and Ke Wang</i> , two talks. Reading random matrices seminar, UCLA
April, 2019	<i>The capacity of feedforward neural networks (based on the paper by Baldi and Vershynin)</i> , two talks. Reading random matrix seminar, UCLA
January, 2019	<i>Uncertainty principle on graphs (based on the paper by Agaskar and Lu)</i> , two talks. Reading random matrices seminar, UCLA
October, 2018	<i>Heavy-tailed random matrices: some non-asymptotic results and questions</i> , Random matrices seminar, UCLA
January, 2017	<i>Structured Random Matrices tutorial</i> , Analysis/Probability learning seminar, University of Michigan
September, 2016	<i>Matrix regularizing effects of Gaussian perturbations</i> , Analysis/Probability learning seminar, University of Michigan
October, 2015	<i>Geometric random walks</i> , Student Analysis, University of Michigan
October, 2015	<i>Refined epsilon-nets and invertibility of random square matrices with i.i.d. heavy-tailed entries</i> , Analysis/Probability learning seminar, University of Michigan
March, 2015	<i>The smallest singular value of random rectangular matrices with no moment assumptions on entries</i> , Analysis/Probability learning seminar, University of Michigan
March, 2015	<i>Small ball probabilities and an introduction to the Littlewood-Offord problem</i> , Student Analysis seminar, University of Michigan
November, 2014	<i>Bounds for the volume of the sections of the unit cube in R^n (by Keith Ball)</i> , Student Analysis seminar, University of Michigan
2014-2015	Series of talks on Geometric analysis reading seminar (topics included: contact points and John's theorem; Milman's quotient of subspace theorem and Bourgain-Milman inequality; Paouris' deviation inequality, Banach-Mazur distance to the cube estimates), University of Michigan

Schools and workshops participation

November, 2019	Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA
May 2018	IPAM Workshop III: Random Matrices and Free Probability Theory within Quantitative Linear Algebra long program, Los Angeles, CA
November, 2017	MSRI program workshop: Geometric functional analysis and applications, Berkeley, CA
August, 2017	Introductory Workshop: phenomena in high dimensions, MSRI, Berkeley, CA
August, 2017	Connections for Women: geometry and probability in high dimensions, MSRI, Berkeley, CA

July, 2016	The Mathematics of Data, Graduate summer school, IAS/Park City Mathematics Institute, Utah
June, 2016	CMS Summer Meeting, University of Edmonton, Alberta, Canada
June, 2016	Summer School on Random Matrices, University of Michigan, Ann Arbor, MI
April, 2016	Informal Analysis Seminar, Kent State University
March, 2016	AMS Sectional Meeting, University of Georgia, Athens, GA
November, 2015	Informal Analysis Seminar, Kent State University
November, 2015	Analytic and Probabilistic Techniques in Modern Convex Geometry, University of Missouri
April, 2015	Workshop on information Theory and Concentration Phenomena, Institute for Mathematics and its Applications, Minneapolis
March, 2015	Informal Analysis Seminar, Kent State University
October, 2014	Informal Analysis Seminar, University of Michigan, Ann Arbor
May, 2014	Random Matrix Theory school, program for Women and Mathematics, IAS, Princeton
May, 2012	The XIX International student, postgraduate and young scientist conference “Lomonosov”, Moscow, Russia
July, 2006-2009	Summer school “Contemporary mathematics”, Dubna, Russia